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E/21/334

EE587- INDIVIDUAL PROJECT

Project Overview

This project implements a hardware-accelerated Sobel Edge Detection Filter on a Xilinx Basys3 FPGA board and displays the output on a monitor via VGA

Software Pipeline (Python)

• **convert_grayscale.py:** Resizes the original input image to 128 × 128 pixels (16,384 bytes) in 8-bit grayscale and exports it as hexadecimal data to input hex.txt.
• **generate_sobel_ref.py:** Calculates the theoretical 3×3 Sobel convolution in software to generate the baseline reference files (output ref hex.txt and output ref.jpg).
• **view_output.py:** Reads the hex array produced by Vivado (output sim hex.txt) and reconstructs it back into a viewable image (output from verilog.jpg).

RTL design and Simulation (Verilog)

• **sobel_filter.v:**
The core receives nine 8-bit pixel values (p00 . . . p22) from a 3×3 neighborhood and calculates horizontal (Gx) and vertical (Gy) gradients:
Gx = (p02 + 2p12 + p22) − (p00 + 2p10 + p20)
Gy = (p20 + 2p21 + p22) − (p00 + 2p01 + p02)
The total magnitude is computed as Magnitude = |Gx| + |Gy|.
An 8-bit saturation check clamps values greater than 255 to 8'hFF

• **tb_sobel.v:**
The testbench loads input hex.txt into memory using $readmemh, iterates across the image grid, feeds surrounding pixel clusters into the core, and writes the output to **output_sim_hex.txt.** Simulation completes at 79, 390 ns.

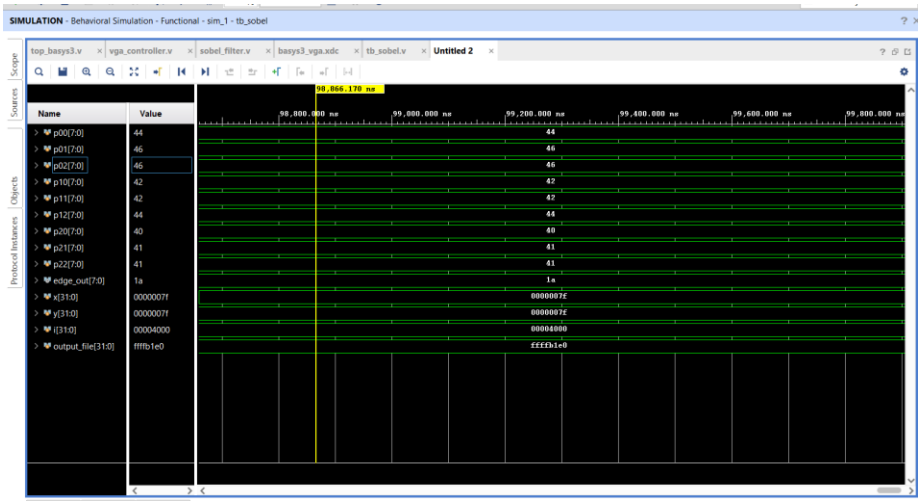


Figure 01: Simulation using Vivado

FPGA Deployment

• **Clock Divider**
Converts the onboard 100 MHz clock to a 25 MHz pixel clock required for standard 640 × 480 @ 60 Hz VGA timing.

• **VGA Controller (vga controller.v)**
Generates horizontal/vertical sync signals (hsync, vsync) and active display coordinates (x, y).

• **Top Module (top basys3.v)**
Pre-loads image output into Block RAM (16 KB), centers the 128 × 128 image on screen (X ∈ [256, 383], Y ∈ [176, 303]), and maps the top 4 bits of pixel data to the 4-bit RGB Basys3 VGA pins.

• **Constraints (basys3 vga.xdc)**
Binds Verilog ports to Basys3 physical pins (Clock, Reset, hsync, vsync, and RGB pins).

Hardware setup

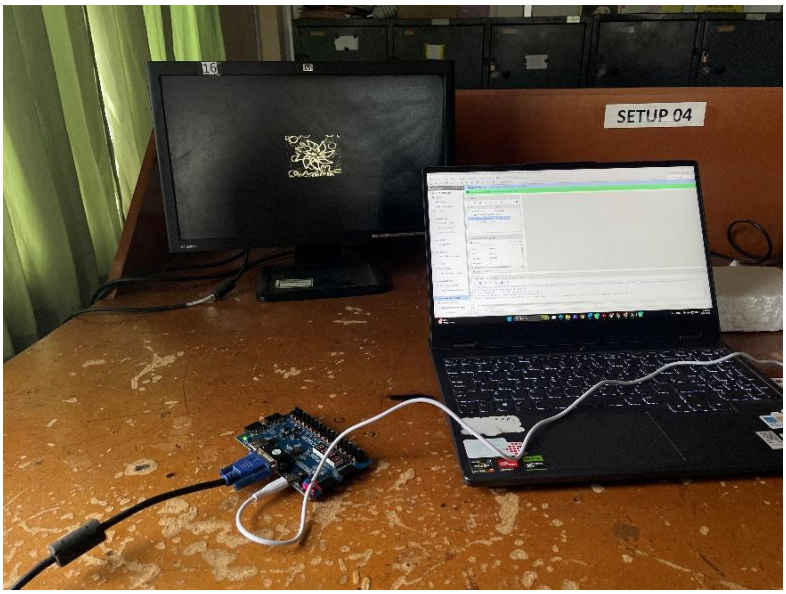


Figure 02: Hardware Setup

Results



Figure 03: Input image



Figure 04: Input image in Greyscale

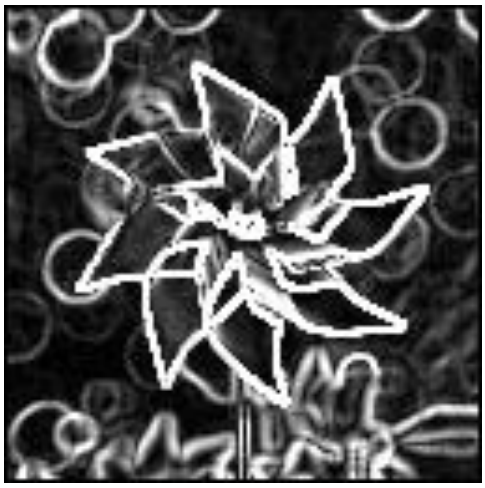


Figure 05: Reference output using python

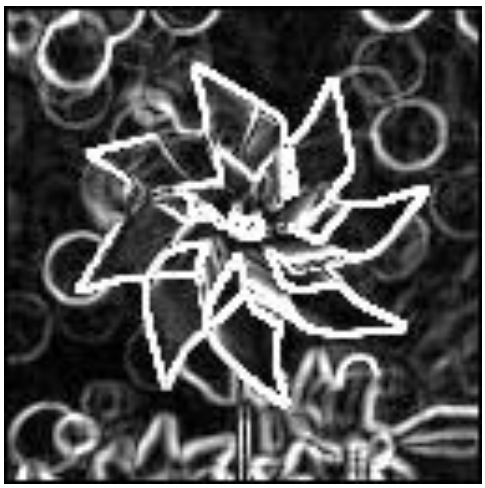


Figure 06: Output from Vivado

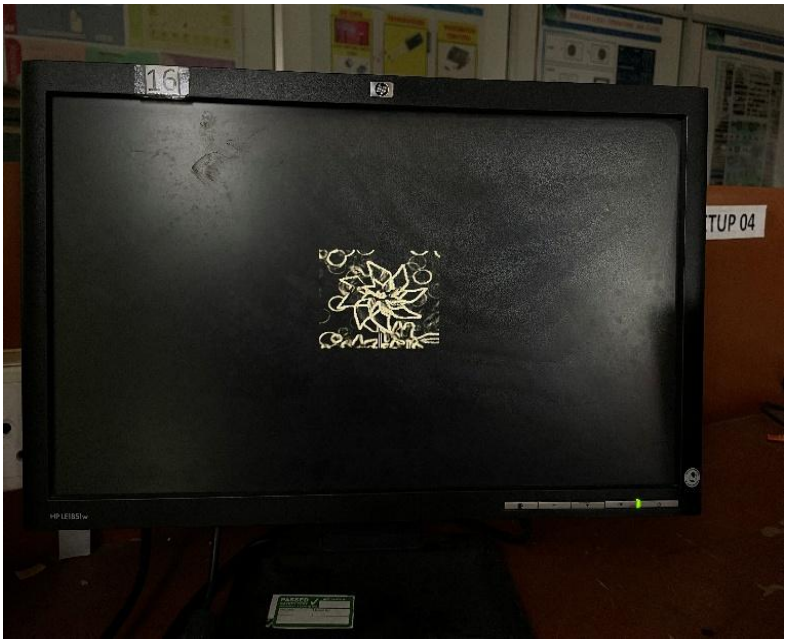


Figure 07: Rendered 128 × 128 edge-detected output image centered on the VGA monitor screen

Summary

Table 1: Summary of Project Source Files

Filename	Type	Function
`convert_grayscale.py`	Python	Resizes image and creates `input_hex.txt`
`generate_sobel_ref.py`	Python	Generates software golden reference
`sobel_filter.v`	Verilog	3×3 Sobel convolution hardware module
`tb_sobel.v`	Verilog	Testbench driving simulation loop
`view_output.py`	Python	Converts simulation hex dump to image
`vga_controller.v`	Verilog	Generates VGA 640×480 sync signals
`top_basys3.v`	Verilog	Top module integrating BRAM, VGA, and clock divider
`basys3_vga.xdc`	XDC Constraints	Pin assignments for Basys3 FPGA board